INCH-POUND

MIL-B-45022F
2 November 1992
SUPERSEDING
MIL-B-45022E
25 October 1985

# MILITARY SPECIFICATION

## BLAST CLEANING CABINETS

This specification is approved for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers liquid and dry abrasive blast cleaning cabinets equipped with manually controlled devices for ejecting abrasive material, under pressure, onto various objects for removal of rust, paint, scale, burrs, etc.

1.2 Classification. The blast cleaning cabinets shall be of the following types, classes, and styles as specified (see 6.2.1).

Type I - Liquid abrasive and compressed air
Type II - Dry abrasive and compressed air

Class 1 - Direct pressure
Class 2 - Suction induction
Class 3 - Gravity induction

Style A - Stationary
Style B - Portable

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| TT-P-664 | - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting |
| UU-T-81 | - Tags, Shipping and Stock |
| PPP-B-566 | - Box, Folding, Paperboard |
| PPP-B-636 | - Boxes, Shipping, Fiberboard |
| PPP-B-676 | - Boxes Setup |
| PPP-C-843 | - Cushioning Material, Cellulosic |
| PPP-T-60 | - Tape: Packaging, Waterproof |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of |
| MIL-B-121 | - Barrier Material, Greaseproofed, Waterproofed, Flexible |
| MIL-T-152 | - Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Associated Electrical Equipment. |
| MIL-V-173 | - Varnish, Moisture-and-Fungus-Resistant (For Treatment of communications, Electronic, and Associated Equipment). |
| MIL-H-775 | - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) And Fittings, Nozzles, and Strainers Packaging Of |
| MIL-A-3816 | - Abrasives and Abrasive Products; Packaging of |
| MIL-M-7298 | - Manuals, Technical: Commercial Equipment |
| MIL-S-12514 | - Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric. |
| MIL-E-16298 | - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of |
| MIL-E-17555 | - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of |
| MIL-C-52950 | - Crates, Wood, Open and Covered |

STANDARDS

FEDERAL



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FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-838 - Lubrication Of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems

MIL-HDBK-267 - Guide for selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.



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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B40.1 - Gauges - Pressure Indicating Dial Type - Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME - Boiler and Pressure Vessel Codes.

(Applications for copies should be addressed to the American Society of Society of Mechanical Engineers, 345 East 47 Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380-91 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators
NEMA ICS 1 - General Standards for Industrial Control and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of



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this document shall takes precedence. Nothing in this document, however, supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The blast cleaning cabinets shall be new and of the manufacturer's current commercially marketed design meeting the requirements specified herein. The cabinets shall be designed to be operated by one operator standing outside the cabinet, with arms extending into the cabinet through openings, and manually directing an abrasive blast against a workpiece within the cabinet.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the cabinet. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-91.

3.2.1.1 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the cabinets shall be graduated in the same system.

3.2.2 Power source. Unless otherwise specified (see 6.2.1), the blast cleaning cabinets shall be designed for operation on 230/460 volt, initially wired for 230 volts, 60 hertz (Hz), three phase current. Control switches and circuits shall operate on internally wired 115 volt, 60 Hz, single phase current.

3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and all other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps



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shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricant during shipment. The manufacturer shall include a warning notice in the instruction manual and on the cabinets to remove the temporary caps or stoppers before start-up or use of the cabinets.

3.2.4.1 Lubrication plate or chart. A corrosion-resistant metallic name plate, with permanently legible characters in the English language, shall be permanently attached to each cabinet in a location visible to the operator. If a chart is furnished, it shall be sealed in a transparent plastic folder or laminated between permanently sealed clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the plate or chart:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification number of lubricant, if applicable.

3.2.5 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. Standard parts shall be used to the maximum extent possible. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.6 Maintainability. Design and assembly of the blast cleaning cabinets, and the placement of adjusting features, shall assure that maintenance of operational capability will be reduced to the simplest form of procedures requiring a minimum of skill. Instructions for disassembly, repair, assembly, and adjustment shall be clean, concise, and definitive. Cabinet design shall permit all maintenance actions to be done using only common tools.

3.2.7 Safety and health requirements. Covers, guards, or other safety devices shall be provided. The devices shall not interfere with operation of the cabinets. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the cabinets, whether or not specifically required herein, shall comply with the requirements of OSHA 2206 that are applicable to the cabinets themselves (see 6.4).



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3.3 Construction. The blast cleaning cabinets shall be furnished complete so they can be used for the operations specified herein. Parts used in fabricating the cabinet shall be new and free of defects. Part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance.

3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Castings and forgings. Castings shall be free from visible blowholes, porosity, hardspots, shrinkage defects, cracks or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.3.3 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.3.4 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.3.5 Electrical construction.

3.3.5.1 Electrical components. Electrical components, including relays, starters, switches, wiring, and their components, shall conform to, and be located in accordance with NEMA ICS 1 and NFPA 70 as applicable.

3.3.5.2 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.



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3.3.5.3 Electric power supply cable. An electrical power supply cable shall be installed on each cabinet. The electrical power supply cable shall be not less than ten feet in length and shall terminate in not less than one inch of stripped tinned wire without a receptacle plug. The electrical power supply cable shall be SJO, ST, or STO and shall include proper sheathing/covering in accordance with the National Electric Code (NFPA 70) and OSHA 2206. The power supply cable shall have a voltage/potential and ampere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, of the blast cleaning cabinets in accordance with NFPA 70.

3.3.5.4 Dielectric strength. The electrical circuits of the blast cleaning cabinets shall withstand, without damage, twice their rated voltage, plus 1,000 volts root mean square (rms) at a frequency of 60 Hz, applied between mutually insulated circuits, and between insulated circuits and ground for a period of one minute.

3.3.5.5 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.4 Performance characteristics.

3.4.1 Capacities. The blast cabinets and supports shall be fabricated to rigidly support all components plus a minimum workload of 150 pounds per square foot (psf) of working area. Minimum working area within the blast cabinet shall be as specified by the procuring activity (see 6.2.1).

3.4.2 Functional characteristics (type I cabinets). The liquid abrasive cabinet shall intermix a slurry and compressed air to produce a liquid blast stream. The liquid abrasive cabinet shall be suited for operation with the following:

(a) Abrasives from 100 through 5,000 mesh.
(b) An external water supply of 35 to 50 pounds per square inch gage (psig).
(c) An external air supply of 80 to 90 psig.
(d) An integral system to enable application of detergent, solvent or inhibitor, entrained with water, at a controlled rate.

3.4.3 Functional characteristics (type II cabinets). The direct pressure, class 1, cabinet shall combine abrasive and compressed air in a mixing area before transfer into the discharge hose. The suction induction, class 2, cabinet shall employ an injector principle to mix abrasive and compressed air upon entering the blast gun nozzle. The gravity induction,



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class 3, cabinet is gravity fed from an overhead abrasive hopper into a moving air stream where it is discharged at a preadjusted air pressure through the blast gun nozzle.

3.5 Component details for liquid abrasive (type I) cabinets.

3.5.1 Blast cabinet. The blast cabinet shall be leak-proof against water and dust. Caulking materials shall not be used in fabrication of cabinets. Cabinet walls shall be of sheet steel not less than Manufacturers Standard Gage No. 14 (0.075 inch). When specified (see 6.2.1), the cabinet interior shall be lined with replaceable plates of material and nominal thickness specified by the procuring activity. Cabinets shall accommodate track rails and loading car when required (see 3.7.1 and 3.7.2).

3.5.1.1 Working ports. Working ports (arm holes) shall be provided through the front wall of the blast cabinet. The ports shall be positioned at a convenient height with a center to center distance of 14 to 22 inches. If round ports are used, they shall be a minimum of seven inches in diameter. If elliptical or rectangular ports are used, the opening shall be a minimum of seven inches wide by ten inches high. Molded rubber, gum rubber, or abrasive resistant neoprene gauntlets shall be secured along the periphery of the working ports and extend inside the cabinet. Gauntlets shall be sized to receive standard, molded rubber, industrial gloves to afford the operator double protection from abrasives.

3.5.1.2 Observation window(s). The blast cabinet shall have an observation window(s) located above the arm ports at a height which permits viewing of the entire working area. The window(s) shall be of shatterproof safety glass, or shatterproof acrylic safety glazing, not less than 1/8 inch thick. The window(s) shall be sealed to absorb shock and prevent leakage. Any special tools required for window replacement shall be furnished. Liquid abrasive cabinets shall have facilities for washing the interior surface of the observation window(s), and means for draining the wash water from the cabinet. When specified (see 6.2.1), means to shield the window(s) from outside light reflections shall be furnished. Shielding devices shall not restrict movement of the operator.

3.5.1.3 Cabinet lighting. The blast cabinet shall employ a lighting system which shall be shielded to prevent system damage during blasting operations. The system shall provide a minimum light intensity of 50 foot-candles at the surface of the work support.

3.5.1.4 Work support. The blast cabinet interior shall be furnished with a removable grate or a perforated steel work support which shall safely support 150 psf.



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3.5.1.5 Cabinet doors. Unless otherwise specified (see 6.2.1), the blast cabinet shall have access doors on both sides adjacent to the operator. Access doors shall provide the maximum area possible without impairing the structural design of the cabinet. Closed access doors shall evidence no leakage during continuous operation of the cabinet. Doors shall operate freely before and after blast cleaning operations. The doors shall contain a safety interlock mechanism to prevent blasting operations when the doors are open.

3.5.2 Hopper-tank. Abrasive slurry shall be contained in a hopper-tank fabricated of sheet steel not less than Manufacturers Standard Gage No. 14 (0.075 inch). The hopper shall be sized to ensure a continuous flow of abrasive at full capacity. The hopper shall be shaped to preclude grit accumulation during any 24 hour period of continuous operation. The hopper and cabinet connection shall be such as to preclude buildup of abrasive.

3.5.3 Slurry circulation system. The slurry circulating system shall suspend 40 percent abrasive by volume. The system shall ensure a continuous flow of abrasive to the blasting gun as well as enough excess flow to prevent clogging or buildup of abrasives. Where design requires, a control valve shall be installed in the gun feed line to control liquid quantity and pressure. Pumps having packing boxes shall be provided with an adjustable packing nut or spring-loaded device to ensure proper sealing of the fluid. Pumps with packing boxes shall have replaceable shafts. The circulating system, when started after a 24 hour shutdown, shall return the entire load of abrasive to complete suspension within five minutes.

3.5.3.1 Circulating system piping. The intake line of the circulating system shall be located above the normal abrasive settling line. All pipe and hose lines shall be located in a corner of the cabinet to provide an obstacle-free working area.

3.5.3.2 Circulating system control. Where required by design, bronze plug-type valves shall be installed in the suction line between the hopper and pump intake and in the abrasive handling line to protect against plugging when the unit is not in operation.

3.5.4 Exhaust system. An exhaust system incorporating a motor, fan, and removable filter shall be furnished to evacuate abrasive vapor from the blast cabinet during the blasting operation. Facilities shall be incorporated to allow flushing the filter with water without opening or removing the housing. The exhaust system shall be sized to prevent a blast cabinet atmosphere which might obscure clear vision of the work being cleaned.

3.5.5 Blast gun and nozzle. A blast gun shall intermix slurry and compressed air, ejecting the mixture onto the workpiece. A foot or leg actuated control (normally off, dead man device) shall regulate the air supply to the blast gun. The blast gun shall be suspended from the top or side of



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the blast cabinet. The suspension system shall allow the gun to reach all portions of the work area. In addition, a suitable bracket shall be provided with the blast cabinet for holding the gun in and out of the way place while loading and unloading workpieces. The blast gun body shall be of corrosion resistant material. At least two styles of gun nozzles shall be available; one producing a round flow for use on heavy workpieces, and one producing a fan-shaped flow for use on light workpieces such as sheet stock. Nozzle styles and sizes shall be as specified by the procuring activity (see 6.2.1). Unless otherwise specified (see 6.2.1), round flow nozzles shall be of boron carbide, tungsten carbide, ceramic material, or of a material of equal or better wear resistance.

3.5.6 Abrasive hose. The abrasive hose shall be a minimum of 1/2 inch inside diameter, abrasive resistant tubing of braided or wrapped construction for use up to 150 psig on class 1, 2, and 3 blast cleaning cabinets.

3.5.7 Air pressure gage. An air pressure gage conforming to ANSI B40.1, grade B, shall be installed on the air intake line. The gage shall indicate a minimum range of 0 to 150 psig air pressure.

3.5.8 Motors. Unless otherwise specified (see 6.2.1), the main motor shall be an induction, squirrel-cage type. Other motors shall be capacitor start. The motors shall conform to the requirements of NEMA MG-1. Speed and torque characteristics shall be adequate to properly operate the cabinet. The motors shall be equipped with starters conforming to type I, class 2 of MIL-S-12514. A push button station, located within reach of the operator, shall be provided for the starter.

3.6 Component details for dry abrasive (type II) cabinet.

3.6.1 Blast cabinet. The requirements of 3.5.1 through 3.5.1.5 shall apply, except that dry abrasive cabinets need not be leak-proof against water. On dry abrasive cabinets, incoming air shall flow across the inside surface of the observation window(s) to aid in dust removal.

3.6.2 Abrasive hopper, direct pressure tank, and mixing area (class 1 cabinet. The abrasive hopper shall be fabricated of sheet steel not less than Manufacturers Standard Gage No. 14 (0.075 inch). The hopper shall be shaped and attached to the blast cabinet so that all spent abrasive falls into the hopper. A screen shall be provided to prevent foreign material that may clog abrasive circulation from entering the hopper. Abrasive shall be returned to the direct pressure tank either directly or through an abrasive reclaimer. The pressure tank shall be of the automatic refilling type, fabricated in accordance with the ASME Boiler and Pressure Vessel Codes (see 2.2), and designed for an operating pressure range of at least 30 to 100 psig. The hopper, tank, and mixing area shall be sized to assure a continuous flow of abrasive at full capacity.



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3.6.3 Abrasive hopper and suction pickup (class 2 cabinet). The requirements of 3.6.2 shall apply, except that a suction pickup system, incorporating means to control and shut off abrasive flow to the blast gun, shall be employed instead of the pressure tank, mixing area, and related check valves. The flow controller shall be adjustable to any given setting with resetting being unnecessary even when the gun blast stream is turned on and off repeatedly.

3.6.4 Abrasive hopper, overhead gravity induction (class 3 cabinet). The class 3 blasting cabinet shall provide a constant supply of abrasive from an overhead hopper, by gravity feed, into a moving air stream mixed at the blast nozzle. The abrasive shall be collected in a hopper located in the bottom of the cabinet and returned to the overhead gravity feed hopper by ducts through an abrasive reclaimer. The overhead hopper shall have means to control abrasive flow to the blast gun.

3.6.5 Piping. Air and abrasive lines shall be located to preclude interference with the blasting operation. The lines shall be accessible for repair and replacement.

3.6.6 Exhauster and abrasive reclaimer. The blast cabinet shall incorporate a suitable exhauster and an abrasive reclaimer that shall separate usable abrasive from unusable dust, and deposit the usable abrasive into the direct pressure tank or the abrasive feed hopper, as applicable. The unusable dust shall be removed from the cabinet by the exhauster. The abrasive reclaimer shall be fabricated of sheet steel not less than Manufacturers Standard Gage No. 14 (0.075 inch). The reclaimer shall be adjustable to permit retention of particles of selected sized and rejection of particles smaller than the selected sizes. Adjustment shall be accomplished without the use of special tools.

3.6.7 Air shut-off valve. A shut-off valve shall be installed in the external air supply line upstream from the air pressure gage.

3.6.8 Blast gun and nozzle. With the direct pressure principle (type II, class 1), the air and abrasive shall be mixed in a mixing area with only one hose attached to the blast nozzle. With the suction or gravity principle (type II, class 2 and class 3), the air and abrasive shall be mixed in the blast gun. This requires two hoses, an air hose and an abrasive hose, to be attached to the gun.

3.6.9 Abrasive hose. The requirements of 3.5.6 shall apply.

3.6.10 Air pressure gage. The requirements of 3.5.7 shall apply.

3.6.11 Motors. The requirements of 3.5.8 shall apply.



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3.6.12 Dust collector. Unless otherwise specified (see 6.2.1), a separate dust collector shall be furnished with type II, class 1 and class 2 cabinets. Unless otherwise specified (see 6.2.1), an integral dust collector shall be furnished with class 3 cabinets and be located on the same framework as the cabinet. The collector shall be dust tight. The collector shall include vertically hung fire resistant cotton cloth filters, a manually operated filter shaking mechanism, and an exhaust system providing a maximum air-to-cloth ratio of 4 to 1 at a static water pressure of six inches. A drawer, pan, or other means shall be provided for removal of collected dust from the bottom of the dust collector. The cotton cloth filters shall be of the bag or tube type, with the seams multiple stitched. Air tight seals shall be provided at the open end of the filter bags.

3.6.13 Carriage (style B cabinet only). Portability shall be attained by use of a wheeled carriage securely attached to the blast cleaning cabinet. The number, rim width, and diameter of the carriage wheels shall be proportioned to allow easy movement over surfaces covered with loose abrasive. Unless otherwise specified (see 6.2.1), wheel rims shall be of iron or steel.

3.7 Accessories. Accessories specified in 3.7.1 through 3.7.4 shall be furnished when so specified by the procuring activity (see 6.2.1).

3.7.1 Loading car track. Track and extension assemblies shall support the loading car of 3.7.2, plus a minimum workload of 150 psf of car loading area. Track shall be provided both inside and outside the blast cabinet. The inbound end of the external track shall be hinged so that it may be folded back upon itself for movement or storage. Unless otherwise specified (see 6.2.1), the length of the external track shall be at least that of the internal track.

3.7.2 Loading car and turntable. The loading car shall have a cast iron or fabricated steel frame, and four grooved or flange type wheels. The car shall roll on the tracks specified in 3.7.1. A bearing-mounted turntable at least 17 inches in diameter shall be installed in the center of the loading car. The car and turntable shall safely withstand a workload of 150 psf of car loading area. The distance from the top of the turntable to the top of car tracks shall not exceed seven inches. Car wheel and turntable bearings shall be sealed against slurry and dust contamination, and shall turn freely when under load and after continued use in the blast cabinet.

3.7.3 Portable tumbling barrel. A portable tumbling barrel shall be mounted on a loading car which rolls on the track specified in 3.7.1. The barrel section shall be perforated to allow drainage of abrasive, and sized to permit its entry through the blast cabinet door. The outboard end of the car shall have a baffle section which seals against the door panel to protect the barrel drive mechanism from the blast within the cabinet. The barrel shall be driven by a motor and gear reducer drive. All bearings exposed to the interior of the blast cabinet shall be protected by proper seals. The



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tumbling barrel shall be furnished with a one to 20 minute timer, a starter and push button station, and a solenoid operated air valve. When specified by the procuring activity (see 6.2.1), a motor driven tumbling barrel, for batch loading of small parts, shall be mounted on the inside of a door panel.

3.7.4 Rinse pan and splash pan (type I cabinet only). The external car track shall be mounted over a sturdy rinse pan of welded construction. The rinse pan shall be shaped to allow collection and drainage of water used in rinsing heavy workpieces. A splash pan shall be bolted around the outboard edge of the rinse pan, and shall rise vertically not less than 18 inches in order to prevent rinse water from splashing into adjoining areas. The rinse and splash pan assembly shall be constructed to allow attachment to the right as well as the left side of the blast cabinet.

3.7.5 Abrasive. When so specified (see 6.2.1), various abrasives shall be furnished with each blast cabinet.

3.7.6 Spare (repair) parts. When so specified (see 6.2.1), spare parts shall be furnished with each blast cabinet.

3.8 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent, i.e., epoxy enamel, shall be applied.

3.9 Information marking.

3.9.1 Nameplate. The nameplate and information plates may be incorporated into one or more corrosion-resistant metallic plates at the manufacturer's option in accordance with MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured to the cabinets with permanent fasteners such as screws, bolts, and rivets at each of the four corners. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number



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3.9.2 Instruction plate. Corrosion resistant metallic instruction and indicating plates shall be permanently attached to the front of the cabinet. The instructions shall be in the English language and shall be permanently and legibly engraved, etched, embossed or stamped in bold face on a contrasting background.

3.10 Workmanship. Standards of workmanship shall assure that the blast cleaning cabinets shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.



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4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

| | Inspection or Test | Lot or batch Size | Sample Size |
|---|---|---|---|
| (a) | Product Examination (4.5) | 1-150 | 13 |
| (b) | Operational Test (4.6.1) | 1-150 | 13 |
| (c) | Functional Test (4.6.2) | 1-150 | 13 |
| (d) | Pressure Test (4.6.3) | 1-150 | 13 |
| (e) | Load Test (4.6.9) | 1-150 | 13 |
| (f) | Dielectric Test (4.6.5) | 1-150 | 5 |
| (g) | Packaging inspection (4.7) | 1-150 | 5 |

4.5 Product examination. Visually, dimensionally, and manually examine the blast cleaning cabinets to determine conformance with the requirements of 3.2 through 3.3.5.3, 3.3.5.5, and 3.5 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the blast cleaning cabinets and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit or function of the end item (see 3.2.5), shall be cause for rejection.

4.6 Tests.



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4.6.1 Operational test. The blast cleaning cabinets shall be cycled through an entire range of operations without the use of abrasives. Proper operation of all controls, motors, and adjusting mechanisms shall be verified.

4.6.2 Functional test. The blast cleaning cabinets shall be operated in accordance with the manufacturer's instructions to remove rust, scale, or paint from the surfaces of irregularly shaped objects. Observations shall be made for the following:

(a) Proper circulation of the abrasive.
(b) Proper operation of blast gun.
(c) Proper exhaust of vapor (type I cabinets).
(d) Proper exhaust of dust and operation of the dust collector (type II cabinets).
(e) Proper operation and fit of access doors.
(f) Window cleaning capability.
(g) Lighting within the cabinet.
(h) Effectiveness of the cleaning operation on the test objects.

4.6.3 Pressure test. All air, abrasive, and water lines shall be subjected to an internal static air pressure of twice the working pressure of the line. While under pressure, the lines shall show no evidence of leakage (see 3.4.2 and 3.4.3).

4.6.4 Load test. The entire working area of the work support shall be subjected to an equally distributed static load of 150 psf. When the loading car and track are supplied, the loading car shall be subject to an equally distributed load of 150 psf on the turntable area and the car rolled on the track into the cabinet. There shall be no deformation or malfunction of components (see 3.4.1, 3.5.1.4, and 3.7.2).

4.6.5 Dielectric test. Twice rated voltage, plus 1,000 volts rms at 60 Hz, shall be applied to mutually insulated circuits and between insulated circuits and ground for a period of one minute to determine compliance with 3.3.5.4.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

## 5 PACKAGING

5.1 Disassembly. Disassembly of the blast cleaning cabinet may be required to reduce cubage, for the protection of parts vulnerable to damage and to assure that all surfaces requiring a preservative are accessible for application. The item shall not be disassembled to the extent that it would require special skills and tools to reassemble.



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5.2 Matchmarking. Removed parts and assemblies shall be matchmarked with mating parts to facilitate reassembly. Matchmarking information shall be printed on tags conforming to UU-T-81, type A, and secured to mating parts and assemblies. The marked tags shall be waterproofed in accordance with MIL-STD-129.

5.3 Preservation/packaging. Preservation/packaging shall be level A or C as specified.

5.3.1 Cleaning and drying. Cleaning shall be in accordance with process C-1 of MIL-P-116 and drying shall be in accordance with any applicable procedure specified therein.

5.3.2 Preservation application. Application of preservatives shall be in accordance with MIL-P-116. Immediately after cleaning and drying all unpainted and non-critical areas shall be coated with type P-1 preservative. The unprimed and unpainted metal components of the exhaust fan assembly, excluding the motor and starter, shall be preserved with type P-2 preservative. Items preserved with type P-2 preservative shall be wrapped with Grade A barrier material conforming to MIL-B-121. Secure barrier material with tape conforming to PPP-T-60, type optional.

5.3.3 Unit packaging. Each complete blast cleaning cabinet shall be individually packaged and all component parts, except those listed below, shall be packaged in accordance with method I of MIL-P-116.

5.3.4 Drive belts. Drive belts shall be removed from the equipment or released from tension. The face or grooves of all ferrous metal pulleys shall be coated with rust inhibitor primer conforming to TT-P-664. Belts removed from the equipment shall be individually packaged in accordance with method III of MIL-P-116.

5.3.5 Electrical equipment. Electric motors, starters, switches, instrument panels, power cords and electrical equipment shall be packaged in accordance with the requirements of MIL-E-16298 and MIL-E-17555, as applicable.

5.3.6 Gages. All gages, if removable, shall be individually wrapped with cushioning material conforming to PPP-C-843, type II, class C, and secured with tape conforming to PPP-T-60, type optional. The cushioned gages shall be placed in a container conforming to PPP-B-566 or PPP-B-676. The container shall be closed in accordance with the appendix to the applicable box specification. When gages are not removable, cushioning material conforming to PPP-C-843, type II, class C, shall be applied over gage and secured with type conforming to PPP-T-60, type optional.

5.3.7 Hoses and fittings. Hoses and fittings shall be packaged in accordance with MIL-H-775.



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5.3.8 Protective equipment. Protective equipment such as aprons, gloves, etc., shall be packaged in accordance with method III of MIL-P-116. Items shall be cushioned to the extent necessary to provide protection from the hazards of shipping and shall be packaged in a fiberboard container conforming to PPP-B-636, W5c. Closure shall be accomplished in accordance with the container specification of the appendix thereto.

5.3.9 Abrasives. Abrasives shall be packaged in accordance with MIL-A-3816.

5.3.10 Technical publications. Publications and assembly instructions issued with the item shall be packaged in accordance with method IC-1 of MIL-P-116.

5.3.11 Consolidated pack. All disassembled parts, technical publications, component parts and accessories shall be adequately cushioned and consolidated and packed in a container conforming to PPP-B-636, W5c. Container shall be closed in accordance with the container specification or the appendix thereto. The packed carton shall be placed inside the unit container and secured in a manner to restrict movement.

5.3.12 Level C. Level C preservation shall be in accordance with the requirements of MIL-STD-1190. The complete blast cleaning cabinet shall be preserved and packaged in a manner to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity for immediate use.

5.4 Packing. Packing shall be level A, B or C as specified.

5.4.1 Level A. The complete unit shall be packed in a container conforming to MIL-C-52950, type V, style B, covered. Consolidated packs shall be located and secured within the exterior container. Blocking, bracing, anchoring and cushioning shall be in accordance with the container specification and MIL-STD-1186. Containers shall be closed and strapped in accordance with the container specification or the appendix thereto.

5.4.2 Level B. The complete unit shall be packed the same as level A requirements.

5.4.3 Level C. Level C packing shall be in accordance with the requirements of MIL-STD-1190. The complete unit shall be packed in a manner which will afford adequate protection against damage during shipment within the Continental United States from the supply source to the first receiving activity.

5.5 Marking. Marking of all interior and exterior containers shall be as specified in accordance with the requirements of MIL-STD-129.



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5.6 Rough handling. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s) specified elsewhere in this documents.

6. NOTES

6.1 Intended use. The blast cleaning cabinets covered by this specification are intended for the following uses:

(a) The type I (liquid abrasive) cabinets are intended to be used for removing rust, paint, scale, and burrs; for matte finishing; for preparation of surfaces for plating; and for cleaning dies, molds, etc.
(b) The type II (dry abrasive) cabinets are intended to be used for removing rust, paint, and scale from metal shapes; for cleaning castings, forgings, gears, etc.; and for decorating masonry, glass, and metal.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2).
(c) First article when required (see 3.1).
(d) Measuring and indicating device graduation(s) required (see 3.2.1.1).
(e) Power source if different (see 3.2.2).
(f) Blast cabinet working area required (see 3.4.1).
(g) Cabinet lining when required (see 3.5.1).
(h) Light reflection shield when required (see 3.5.1.2).
(i) Cabinet doors if different (see 3.5.1.5).
(j) Nozzle styles and sizes as required (see 3.5.5 and 3.6.8).
(k) Nozzle fabrication if different (see 3.5.5 and 3.6.8).
(l) Main motor if different (see 3.5.8).
(m) Dust collector for type II, classes 1 and 2, if different (see 3.6.12).
(n) Integral dust collector for type II, class 3, if different (see 3.6.12).
(o) Wheel rims if different (see 3.6.13).
(p) Accessories required (see 3.7).
(q) External track length if different (see 3.7.1).
(r) Tumbling barrel for door mounting when required (see 3.7.3).



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(s) Abrasive when required (see 3.7.5).
(t) Spare parts when required (see 3.7.6).
(u) Painting and finishing if different (see 3.8).
(v) First article inspection if different (see 4.3).
(w) Quality conformance inspection if different (see 4.4).
(x) Level of preservation/packaging and packing required (see 5.3 and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Standard lubricants. The cabinet manufacturer should specify and use to the maximum extent possible those lubricants (see 3.2.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - AS

Preparing activity:

Army - AL

Project No. 4940-0626



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | | |

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION |
|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) / e. DATE SUBMITTED (YYMMDD) |

7. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) (1) Commercial / (2) DSN |
|---|---|
| Cmdr USA Armament Res, Dev. & Engr Ctr | (319) 782-8146 / 793-8146 |
| c. ADDRESS (Include Zip Code) ATIN: SMCAR-EST-S Rock Island, IL 61299-7300 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340 |

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